

Why is everyone acting like AI using more energy or water is automatically bad?

Posted by u/johnnybgoat • 0 points • 12 comments

Okay so, hear me out for a second.

All this is doing is exposing how horrifically outdated our energy grid already is. Civilization has always advanced by using more energy, not less. We were bound to hit this cap sooner or later.

Coal is dirty. Renewables help, but they don't run 24/7 and require huge land areas and lots of materials. People keep saying "just build more solar and wind instead of nuclear" but solar only works when the sun is out and wind only works when it's windy. To match a single nuclear plant, you often need thousands of turbines or massive solar fields, plus backup power for when they stop producing.

And nobody talks about the material cost neither for some reason. Solar and wind need huge amounts of steel, concrete, aluminum, rare earth metals, and copper. Their parts wear out every 15 to 25 years. A nuclear plant uses far fewer materials overall relatively speaking because one reactor matches the output of massive renewable fields and it lasts two to three times longer.

AI increasing energy demand actually creates a positive loop objectively speaking as it forces countries to modernize the grid, move away from coal and oil, and build cleaner long-term power systems thus leading to more stable and cleaner energy overall.

As for water... yes, AI uses water for cooling, closed loop and water cycle aside for a second... once the grid is upgraded anyway, we can literally finally scale up desalination, producing more clean drinking water. We have the technology for "unlimited" fresh water. Energy was always the bottleneck and this will solve it magnificently.

Yes, the transition sucks. Prices are rising everywhere. Personally I hate it too. But long-

term, this looks more like fixing problems we've ignored for decades than some doom scenario.

P.s: for those mentioning nuclear waste, more advanced AI beyond our current primitive LLM will most likely allow us to deal with it better. But overall, I'd say it's better to have a localized nuclear dump rather than pumping it straight into the air with coal for everyone to enjoy worldwide.

Comments

u/rasta4200 • 1 points • 2026-03-28 20:12 UTC

In realtà l'ipotesi più plausibile è che l'IA si produca l'energia da sola dopo aver acquistato i reattori nucleari già esistenti e fermi, permettendo alla rete civile di progredire con le rinnovabili. Anche a intuito, un gigante dell'AI che mette server su server ha come unica limitazione l'elettricità, e si sa che i progetti statali sono più lunghi di quelli privati. E infatti tutte le aziende AI sono attive nella ricerca della fusione nucleare, degli SMR e hanno già finanziato progetti di costruzione di nuove centrali nucleari dedicate a loro, rimangono i dubbi sulle scorie e sull'approvvigionamento di minerali per il nucleare, anche se al momento Trump ha fermato i depositi sotterranei di combustibile esausto per riutilizzarlo in nuove centrali nucleari avanzate, e comunque non abbiamo scelta, o il nucleare o la capacità di respirare, la quantità di gas naturale necessario per reggere l'IA produrrebbe tanta CO₂ da distruggerci completamente. Purtroppo Trump ha eliminato il limite di emissioni nelle centrali a gas, e con i dazi non è assicurata il rifornimento di minerali, il che porta l'America a una risalita momentanea ma a una discesa vertiginosa nei prossimi 30 anni, tutto per inseguire la Cina, che intanto domina sia con i reattori in costruzione, sia con le rinnovabili. L'Unione Europea in tutto ciò sta in mezzo a fare da vigile urbano e da regolamentatore, usando il potere d'acquisto dei cittadini europei (importantissimo per la Cina) per creare standard per l'IA e nuove tecnologie e per farle standardizzare in tutto il mondo, impedendo a un'America senza un capo di distruggere il resto del mondo.

u/rasta4200 • 1 points • 2026-03-28 20:16 UTC

Tra l'altro per i server l'IA le rinnovabili non sono adatte perché per l'IA serve continuità giorno e notte, mentre per i cittadini e le aziende sì, obbligando i governi a costruire più rinnovabili possibili, visto che il gas verrà dirottato principalmente nella produzione di energia per i server l'IA nel momento di picchi di consumo, e soprattutto grazie anche alle politiche estere russe e iraniane, ora i nostri governi sono molto meno contenti di usare il gas.

u/rasta4200 • 1 points • 2026-03-28 20:16 UTC

"politiche estere"

u/admiralackbar2019 • 1 points • 2026-02-23 17:30 UTC

Because none of this "AI" garbage has improved life by 1% in any measurable way

u/WhyAreYallFascists • 1 points • 2026-02-23 16:42 UTC

Closed loop systems would be cool if they were using them.

u/Human_certified • 1 points • 2026-02-22 13:14 UTC

Suppose electricity really was constrained (in the long run it isn't, in the short run it's limited by time-to-build).

Suppose water use really was constrained (it really isn't at all, and water use is not intrinsic to AI).

Then we would have to make hard choices: Which industries stay? Which industries go and shut down?

And you'd get the RAM situation all over:

AI wins, because AI is more valuable, AI is more efficient, there is more demand for AI, there is more growth potential for AI.

AI doesn't happen *in addition* to other industries, and has to "beg for scraps" because it came last.

AI would actually *crowd out* other industries, because it comes first. Markets have been very clear about this.

u/Plastic_Bottle1014 • 2 points • 2026-02-22 08:12 UTC

Yeah, the energy debate has really polluted the conversation. Most people don't really understand when, why, and how different energy sources are bad.

The issue with AI and energy comes down to how much it uses and how much heat it creates all at once. On one hand, it's more efficient. Scientists generating a protein prediction means that's decades worth of energy saved by not needing to out in decades of work. It's more efficient, and is overall less energy. However, these scientists aren't going to retire now. They're going to continue researching and using AI to advance fields. In that regard, we now use energy at a significantly faster rate than before. While it may be beneficial for human advancement, we don't have great systems in place for countering this energy use.

It becomes the same for AI used in hobbies. Something like digital art already caused pollution. However, it wasn't discussed because it wasn't creating so much that it couldn't be offset. On the other hand, it was easy to ignore because blame was shifted to power companies while simultaneously pulled away from the people that were demanding that power.

My point is: this is a different scale scenario. It's fast and hard vs slow and sustainable. We cause more greenhouse gasses and heat generation than ever, but emissions regulations and cleaner fuel has been able to keep us relatively stable. Meanwhile the most significant damage to the Ozone was done by rapid usage of CFCs because they were pumping into the atmosphere way faster than they could break down and become harmless.

If we expanded cleaner and more efficient energy sources, then this would be less of a concern or issue. At the same token, I don't think AI is going to keep expanding at the current alarming rate. It's not like every household is going to have an AI data center.

u/Most-Ad4680 • 1 points • 2026-02-22 04:05 UTC

Why is everyone acting like serial killers murdering multiple people is bad? All their doing is exposing the state of policing in America and the abysmal closure rate on homicides in this country. Really, I think its great more people are ending up dead, because that will somehow in some way make it so less

people end up dead or something, I don't really know how to read and just have gemini post on reddit for me

u/KFrancesC • 2 points • 2026-02-22 02:17 UTC

You think a Republican administration is going to upgrade any type of infrastructure? That cost LOTS of tax dollars!

Because of course, corporations aren't expected to upgrade private infrastructure they own, that falls on tax payers.

And Republicans are too busy giving away all our money to those Corporations, who won't upgrade our infrastructure, and we have none left for anything else.

So either way this problem, and it will eventually cause problems, will be left for all of us to pay for, out of our pockets. Making us poorer

While corporations will make billions of AI tech.

So I think all that is a decent reason, to believe this is all a problem.

u/rasta4200 • 1 points • 2026-03-28 20:21 UTC

I hope that you guys start to change the way you vote for president change. I think that trump should be the last that benefits from the holes in the laws.

u/Distinct-Raspberry21 • 1 points • 2026-02-22 00:00 UTC

If the grid is an issue then we should be working on making the grid better, not use the shitty grid we have to power data centers. What even is your argument here, cause like the AI companies aren't power companies, and we been paying them for decades and they've only been shitty, someone giving them more money than us is going to make them shittier to us as people have never been their focus. They have had to have court orders put in place to not turn off power during heat/cold waves.

u/BorgsCube • 1 points • 2026-02-21 23:26 UTC

AI aside we should've been investing so much more into nuclear, but a lot of communities just don't want them around, nimbys I think is what it's called? (not in my back yard)

u/Ambitious-Wind9838 • 1 points • 2026-02-22 07:11 UTC

Nuclear energy is absolutely useless for the AI race. Energy is needed yesterday, and nuclear power plants are built insanely slowly.

u/BorgsCube • 1 points • 2026-02-22 07:13 UTC

all the more reason to start now

u/Ambitious-Wind9838 • 1 points • 2026-02-22 08:11 UTC

The problem is that AI makes the future extremely uncertain. What will the situation be like tomorrow? Perhaps AI won't be able to meet expectations, and energy demand will fall catastrophically, in which case nuclear power plants will become an unnecessary burden. Perhaps scientists will optimize AI, requiring radically less energy to maintain, which will make nuclear power plants obsolete again. Or perhaps we will create superintelligence, and super AI will provide us with incredibly cheap fusion energy, making all nuclear power plants obsolete?

Therefore, companies prefer to act tactically rather than strategically due to the uncertainty of the future.

u/skr_replicator • 1 points • 2026-02-21 23:16 UTC

If I'm not mistaken, Solar has progressed so much that it gives you more energy per construction investment, and is built like 20x faster. And while it doesn't work in the dark, there are batteries for that (and we really need to finally figure out some advancement in this field). And wind+solar don't share the same blind spot, so one can carry it while the other is blinded.

u/rasta4200 • 1 points • 2026-03-28 20:27 UTC

Tra l'altro ho fatto la tesi sul solare, e una tecnologia relativamente nuova e sempre in sviluppo, ma ha un problema: ha bisogno di terre rare lei stessa e inquina il processo di costruzione del pannello, che dopo 15-20 anni andrà sostituito, e in una società spaventata dal riciclo questo non è sostenibile almeno quanto bruciare gas, a lungo termine.

u/rasta4200 • 1 points • 2026-03-28 20:23 UTC

Gia, ma non produrrebbe abbastanza per soddisfare la crescita dell'AI. Per quello il nucleare è l'unica risposta. E tranquilli, oltre a riaprire Three Mile Island stanno costruendo le loro

u/rasta4200 • 1 points • 2026-03-28 20:25 UTC

Se il tranquilli era un po' ironico, anche se ormai il nucleare non rappresenta più una minaccia ma solo un problema per gli stati senza una legge per lo smaltimento dei rifiuti radioattivi (da quello che ho letto nello stato della centrale di Three Mile Island non c'è nessuna legge) ma potenzialmente questa è l'unica soluzione all'AI e alle bollette alte per chi gli sta accanto.

u/The_Webweaver • 1 points • 2026-02-23 15:35 UTC

My argument has been for years that nuclear+solar+wind is pretty much the ideal energy mix. I don't like hydro for how it interferes with local ecologies and how easy it is for politicians to forget to maintain, and related consequences.

u/skr_replicator • 1 points • 2026-02-23 15:58 UTC

I also think that these 3 are the best trio for now. Just saying that we should not underestimate the advancements in Solar power, how powerful and cheap it's getting, making it a better idea for solar to be the biggest one of these 3, even despite its flaws like nights and dark days.

Building Solar that can be operational in 1 year instead of 20, and deliver as much wattage per investment as the same investment in nuclear, should be preferable to do first.

Let's not forget we already have a solid infrastructure of non-renewable energy. The first investments in solar will just be the fastest and strongest way to quickly reduce our usage of 100% fossils to 20% fossils (mostly as backup during dark days and nights). And then we should build the nuclear to take this remaining backup 20% job of those fossils.

u/FlashyNeedleworker66 • 1 points • 2026-02-21 23:14 UTC

My 2 cents is get in and help if you can. Getting solar was a great decision for me, and I live in "data center alley" while paying like \$10/mo to the utility.

u/bread_eater_0 • 1 points • 2026-02-21 23:11 UTC

Ignoring like the majority of the post BUT coal is actually alot more awful because its ALOT less efficient than even natural gas that america has an abundance of, its not about the efficiency or good of the people, just people with money wanting to make more of it regardless of the means.